

MedVerify AI

An Explainable Medical Claim Verification System Using Hybrid Retrieval-Augmented Generation (RAG), Medical NLP, and Evidence Ranking

1. Abstract

Medical misinformation has become one of the major public health challenges due to the widespread use of social media platforms such as WhatsApp, YouTube, Facebook, Instagram, and X (Twitter). False claims regarding diseases, medicines, vaccines, and treatments often influence people's healthcare decisions, leading to delayed diagnosis, self-medication, and increased health risks.

Current AI-based medical misinformation detection systems mainly classify claims as true or false without providing sufficient evidence, explainability, or updated medical knowledge. Large Language Models (LLMs) can generate helpful responses but may also produce unsupported statements if they are not grounded in reliable sources.

MedVerify AI is proposed as an Explainable Medical Claim Verification Platform that combines Medical NLP, Hybrid Retrieval-Augmented Generation (Hybrid RAG), trusted medical knowledge bases, and evidence-ranking algorithms to verify medical claims. The system retrieves evidence from trusted medical organizations such as WHO, PubMed, CDC, ICMR, NHS, and PubMed Central before generating a response. Instead of providing only a prediction, the system generates an evidence-backed explanation, credibility score, supporting references, and consensus analysis.

The proposed architecture aims to improve transparency, reliability, and trustworthiness while continuously updating its medical knowledge using newly published research papers.

2. Introduction

The internet has made healthcare information easily accessible to everyone. Unfortunately, false medical information spreads much faster than verified information.

Examples include:

- Garlic cures diabetes.
- Vaccines cause autism.
- Drinking warm water kills viruses.
- Cancer can be cured without medical treatment.

People frequently believe these claims because they appear convincing and are shared by friends, influencers, or social media.

Traditional misinformation detection systems simply classify content as true or false. Users receive no explanation and cannot verify why the AI made its decision.

The proposed system solves this problem by verifying every medical claim against trusted healthcare resources before producing a response.

3. Problem Statement

Existing medical misinformation detection systems suffer from several limitations:

- Static knowledge after training.
- Limited explainability.
- No trusted evidence retrieval.
- No credibility measurement.
- Poor handling of conflicting medical research.
- Limited support for multimodal inputs.
- No continuous knowledge updates.

As medical research evolves daily, relying only on pre-trained AI models may result in outdated or incomplete information.

4. Objectives

Primary Objective

Develop an explainable AI platform that verifies medical claims using trusted healthcare knowledge sources.

Specific Objectives

- Detect medical misinformation.
- Extract claims from text, images, and videos.
- Classify disease categories.
- Retrieve evidence from trusted sources.
- Rank evidence based on quality.
- Calculate credibility scores.
- Generate explainable responses.
- Continuously update medical knowledge.
- Support multilingual verification.
- Build an intuitive web dashboard.

5. Scope

The prototype focuses on:

- Diabetes
- Cancer
- Cardiovascular Diseases
- Vaccination
- Nutrition
- Infectious Diseases

Future versions can include additional disease categories.

6. Existing System

Current approaches generally follow:

User Input

↓

BERT

↓

True / False

Limitations

- Black-box predictions
 - No evidence
 - No references
 - Cannot verify latest research
 - Poor explainability
 - Static model knowledge
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7. Proposed System

The proposed system combines multiple AI technologies.

Major Components

- Claim Extraction
- Disease Classification
- Hybrid RAG
- Semantic Search
- Evidence Ranking
- Medical Consensus Analysis
- Explainable AI
- Credibility Scoring

Workflow

User submits claim



Claim Extraction



Disease Classification



Hybrid Retrieval



Evidence Ranking



Medical Consensus



LLM Explanation



Credibility Score



Dashboard

8. System Architecture

Input Layer

Supports

- Text
- Images
- Videos

Image

↓

OCR

↓

Text

Video

↓

Speech-to-Text

↓

Text

↓

Claim Extraction

↓

Disease Classification

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Hybrid Retrieval

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Vector Database

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Live PubMed API

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Evidence Ranking

↓

Consensus Analysis

↓

LLM

↓

Final Verification

↓

Dashboard

9. Disease Categories

- Diabetes
 - Cancer
 - Cardiovascular Diseases
 - Vaccination
 - Infectious Diseases
 - Nutrition
 - Mental Health
 - Kidney Diseases
 - Liver Diseases
 - Women's Health
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10. AI Models

Claim Extraction

Recommended

- Qwen 3
- Llama 3.1
- Gemma 3

Purpose

Extract the medical claim from user input.

Disease Classification

Recommended

- BioBERT

Alternative

- PubMedBERT
- ClinicalBERT

Purpose

Detect the medical domain.

Example

Input

Garlic cures diabetes.

Output

Disease

Diabetes

Embedding Model

Recommended

BAAI/bge-large-en-v1.5

Alternative

MiniLM

Purpose

Convert documents into vectors.

LLM

Recommended

- Qwen 3
- Llama 3.1
- Gemma 3

Purpose

Generate evidence-based explanations.

11. Data Sources

Primary Sources

WHO

- Diabetes Guidelines
- Cancer Guidelines
- Vaccination Guidelines

PubMed

- Latest research papers

PubMed Central

- Open-access full papers

CDC

- Disease prevention

ICMR

- Indian guidelines

NHS

- Clinical recommendations

Training Datasets

- PubHealth
- SciFact
- CoAID
- HealthFact
- MedNLI

12. Database Design

PostgreSQL

Tables

Users

Claims

Verification History

Disease Categories

Feedback

Reports

Vector Database

Stores

Document Chunk

Embedding

Source

Publication Date

Disease Category

Evidence Type

Confidence

13. Hybrid RAG

Instead of searching only one database,

MedVerify combines

Static Knowledge

- WHO
- CDC
- ICMR
- NHS
- Previously indexed PubMed

+

Live Retrieval

- PubMed API

Advantages

- Faster
 - More reliable
 - Up-to-date
-

14. Evidence Ranking Engine

Every retrieved document receives a score.

Example

WHO Guideline

98

Systematic Review

95

Randomized Clinical Trial

90

Meta-analysis

88

Observational Study

75

Case Report

60

Blog

0

This prevents weaker evidence from outweighing trusted medical guidance.

15. Medical Consensus Engine

Instead of selecting one paper,

The system analyzes all retrieved evidence.

Example

Supporting Studies

12

Contradicting Studies

3

Neutral

5

Consensus

Most evidence does not support the claim.

16. Credibility Score

The score combines multiple factors.

- Source reliability
- Evidence agreement
- Publication recency
- Number of supporting studies
- Disease-specific confidence

Example

Credibility

89%

High Confidence

17. Explainable AI

Instead of

Fake

The system shows

Reason

Supporting evidence

Official guideline

Scientific explanation

References

Confidence

18. Automatic Knowledge Updating

A scheduled process periodically:

- Checks PubMed
- Downloads new abstracts or metadata
- Generates embeddings
- Updates the vector database

This keeps the knowledge base current without retraining the entire model.

19. Prototype Modules

Module 1

Authentication

Module 2

Medical Claim Submission

Module 3

OCR

Module 4

Speech-to-Text

Module 5

Claim Extraction

Module 6

Disease Classification

Module 7

Hybrid Retrieval

Module 8

Evidence Ranking

Module 9

Consensus Analysis

Module 10

Explainable AI

Module 11

Dashboard

Module 12

Admin Panel

20. User Interface Prototype

Home



Submit Claim



Processing



Results

Result Page

Medical Claim

Disease Category

Credibility Score

Evidence Summary

Medical Consensus

References

Supporting Studies

Related Guidelines

Download Report

21. Technology Stack

Frontend

React

Tailwind CSS

Backend

Spring Boot

AI Service

FastAPI

AI Models

BioBERT

BGE Embeddings

Qwen 3 / Llama 3.1

Database

PostgreSQL

Vector Database

FAISS

RAG

LangChain

OCR

Tesseract

Speech-to-Text

Whisper

Deployment

Docker

Kubernetes

22. Expected Output

Input

Garlic cures diabetes.

Output

Disease Category

Diabetes

Credibility Score

21%

Evidence

WHO

No recommendation.

PubMed

Current evidence is insufficient.

Medical Consensus

Majority of studies do not support the claim.

Explanation

There is currently no high-quality clinical evidence demonstrating that garlic cures diabetes.

References

WHO

PubMed

CDC

23. Novel Contributions

- Hybrid RAG combining static trusted knowledge with live PubMed retrieval.
 - Disease-aware semantic retrieval.
 - Evidence quality ranking.
 - Medical consensus analysis.
 - Explainable AI with references.
 - Credibility scoring algorithm.
 - Continuous knowledge updates.
 - Multimodal claim verification.
 - Analytics dashboard for misinformation trends.
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24. Future Enhancements

- Browser extension.
- Mobile application.
- Hospital integration.
- Voice assistant.
- Regional guideline support.
- Personalized health recommendations.

- Multilingual verification.
 - AI agent collaboration.
 - Video claim verification.
 - Real-time misinformation monitoring.
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25. Expected Impact

The proposed system helps users verify health-related claims before making medical decisions. Unlike traditional misinformation detectors or generic AI chatbots, MedVerify emphasizes transparent, evidence-backed verification using trusted medical resources and explainable reasoning. This approach can support students, researchers, healthcare professionals, and the general public by improving trust in AI-assisted medical information while reducing the spread of misinformation.